

Algebra 1

Unit 13

Lesson 9 Homework

Name_____

Date_____

Period_____

To make a 15% acid solution, a chemist mixes 3 parts of 4% acid solution with a 20% acid solution. The expression $3(0.04x) + 0.20y$ represents the mixture, where x is the number of ounces of 4% solution and y is the number of ounces of the 20% solution.

Explain what each quantity means.

1. $(0.04x)$

2. $3(0.04x)$

3. $0.20y$

Since 2015, the number of independently contracted food delivery drivers can be modeled by the equation $y = 1,250,000(1.073)^x$.

4. What does the x variable represent?

5. What does the y variable represent?

6. What does the quantity 1,250,000 represent?

7. What is the constant percent rate of change?

The equation $y = -2(x - 4)^2 + 45$ models the height, y , in feet, of a ball, x seconds after it is thrown into the air.

Explain what each quantity represents.

8. $x - 4$

9. 45

Jovi is buying cereal for \$3.25 per box. He has a coupon for two free cereals. He pays \$4.75 for a gallon of milk. The expression $3.25(x - 2) + 4.75$ can be used to represent the overall cost.

10. What does the quantity $3.25(x - 2)$ represent?